Highlights

BowTie Risk Analysis of the Valve Train in a High-Performance VW AP 1.8 Cylinder Head Operating at 10,000 rpm

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- BowTie applied to valve-train risk in a 10 000 rpm AP 1.8 cylinder head
- Top event: loss of valve–cam follower contact (float, bounce, coil bind)
- Quantitative coupling: natural frequency, coil-bind margin, fatigue safety
- Critical preventive barrier identified for the AP 1.8 operating envelope
- BowTie shown to complement FMEA/RPN for high-rpm reliability assessment

BowTie Risk Analysis of the Valve Train in a High-Performance VW AP 1.8 Cylinder Head Operating at 10,000 rpm

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ARTICLE INFO

Keywords:
BowTie analysis
Valve train
Risk assessment
Internal combustion engine
Mechanical reliability

Abstract

The valve train is a load-critical subsystem when a spark-ignition engine is pushed to very high rotational speeds. At 10 000 rpm the inertial demand on the valve–cam contact, the spring dynamics, and the thermal management of the head become tightly coupled, and small departures from the design envelope can cascade into catastrophic events such as valve–piston collision. This work applies the BowTie methodology to the valve train of a high-performance cylinder head conceived for the Volkswagen AP 1.8 platform operating at 10 000 rpm. Threats, preventive barriers, consequences, and mitigative barriers are mapped around the top event "loss of follower contact between valve and cam". The qualitative diagram is then coupled to a quantitative layer derived from the analytical sizing of the intake valve and the dual Cr–Si valve springs, yielding pass/fail margins for the dynamic follower-contact ratio, the spring natural frequency against cam harmonics, the coil-bind clearance, and the fatigue factor of safety. For the AP 1.8 head, all four dynamic-ratio indicators meet the project criterion ($R \geq 1.20$, with the worst case at $R_{\mathrm{eq},-,\mathrm{esc}} = 1.343$), but the analysis identifies two concurrent at-risk barriers in the baseline configuration: (i) the outer-spring coil-bind margin, with $M_{\mathrm{out}} = 0.78$ mm against the project specification of $M \geq 1.5$ mm, and (ii) the inner-spring fatigue Goodman factor of safety, $n_{f,\mathrm{in}} \approx 0.68$ against $n_f \geq 1$ for the fluctuating shear loading between seat preload and full-lift force. The outer-spring natural-frequency margin emerges as a residual concern ($f_{n,\mathrm{out}}/f_{\mathrm{cam,max}} \approx 3.81$). The two critical findings affect different physical components and require independent design interventions, demonstrating the value of the BowTie framework over a single-score FMEA/RPN approach for reliability-aware design of high-rpm cylinder heads.

1. Introduction

Spark-ignition (SI) engines remain technologically relevant in light automotive applications, particularly in markets where cost pressure, mechanical robustness, an established refuelling network, and a gradual transition to lower-impact energy solutions coexist (Heywood, 1988; Stone, 2012). Within this landscape, the search for higher specific output and thermal efficiency does not depend solely on electronic calibration or fuel selection: the geometric quality of the intake system, of the combustion chamber, and of the valve train, together with thermal management of the head, governs the practical envelope of operation.

In high-performance configurations, where elevated rotational speed, increased thermal loading, and stricter operational stability are required simultaneously, two risks acquire a critical role: abnormal combustion (knock) and valve-train failure. Both phenomena compromise reliability, shrink the operational safety margin, and may evolve from a local component issue into catastrophic damage to the engine. The present work focuses exclusively on the second of these two risks. The companion problem of knock at high compression ratio is addressed elsewhere and is out of scope here.

The Volkswagen AP 1.8 platform was adopted as the reference engine for two reasons. First, it occupies an established position in the Brazilian automotive market, with a wide installed base and broad availability of geometric data (Porto, 2026, § 1.6). Second, in the context of a master's research project investigating a high-performance cylinder head for the same platform, a specific operating regime of 10 000 rpm was adopted as the design target. At that rotational speed the inertial demand placed on the valve–cam contact pair is no longer a local component concern: the dynamic stability of the moving valve mass, the residual coil-bind clearance, the natural frequency of the spring relative to cam harmonics, and the fatigue resistance of the spring all become coupled, and a single insufficient margin can propagate into a valve–piston collision.

The conventional Failure Mode and Effects Analysis (FMEA) approach addresses such situations by ranking failure modes through the multiplicative Risk Priority Number (RPN). This approach has known limitations: subjectivity in the assignment of severity, occurrence, and detectability scales; loss of resolution when multiple modes share a similar RPN; and limited visibility of the causal chain that links threats to consequences via barriers. The BowTie methodology addresses these limitations by organising the analysis around a top event, with threats and preventive barriers on the left side, and consequences and mitigative barriers on the right side. This structure makes the role of each barrier explicit and supports communication between design, operation, and reliability domains.

This paper proposes a quantitative-coupled BowTie analysis of the valve train of a high-performance cylinder head operating at 10 000 rpm in the VW AP 1.8 engine. The qualitative bow diagram is enriched with a quantitative layer

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derived from the analytical sizing of the intake valve and of the valve springs. Each preventive barrier is associated with a quantitative pass criterion (natural-frequency margin, coil-bind margin, fatigue factor of safety) so that the BowTie diagram becomes both a communication artefact and a design check.

The contributions of this paper are summarised as follows:

1. A BowTie diagram of the valve train of the AP 1.8 cylinder head at 10 000 rpm, with the top event "loss of follower contact between valve and cam" and the associated set of threats, preventive barriers, consequences, and mitigative barriers.
2. A quantitative coupling layer that translates each preventive barrier into a measurable indicator with a literature-anchored pass criterion, derived from the analytical sizing of the valve and the spring.
3. Identification of the critical preventive barrier for the operating envelope adopted, with a comparison against a conventional FMEA/RPN ranking of the same failure modes.
4. A reproducible methodology that can be transposed to other high-performance cylinder heads operating at high rotational speed.

Novelty statement. The contribution of this work fits the scope of *Engineering Failure Analysis* along four dimensions. First, the manuscript identifies the failure mechanisms of a high-rpm valve train (valve float, bounce, approach to coil bind, surge resonance, fatigue) and proposes preventive actions through a barrier-centric formulation, in line with the journal's declared aims. Second, the work is suitable for the journal because it organises analytical sizing of the valve and the springs around the BowTie methodology, providing a reproducible application of the framework to a specific component of a spark-ignition cylinder head operating at 10 000 rpm. Third, the novelty in relation to the international literature is domain-specific and incremental, not methodological in the broad sense: quantitative extensions of the BowTie diagram have been previously proposed (Khakzad et al., 2013, via Bayesian-network mapping; LOPA, fault-tree-coupled BowTie, and DyPASI in process safety), and the present work does not claim to introduce a new general method. The original contribution is the explicit mapping of each preventive barrier to a deterministic indicator derived from the engineering sizing of the intake valve and the dual valve springs of the AP 1.8 head, with a literature-anchored pass criterion for each, and the identification of the resulting critical barriers under the operating envelope adopted. Fourth, the manuscript relates directly to recent papers published in *Engineering Failure Analysis* on engine-component reliability (Vélez Godiño et al., 2019; Soffritti et al., 2018; Xing et al., 2021; Zhao et al., 2023) by extending their post-failure case-study findings into a structured prospective barrier-evaluation procedure applicable at the design stage. The claim that no published study applies the BowTie methodology to the valve train of a high-rpm spark-ignition cylinder head is supported by a search of the Scopus and Web of Science databases conducted in April 2026 with the query strings `("BowTie" OR "bow-tie") AND ("valve train" OR "valve-train" OR "valve spring") AND ("spark ignition" OR "SI engine" OR "high rpm")`, which returned no relevant hits.

Section 2 reviews the relevant failure modes of the valve train at high rpm and the BowTie methodology. Section 3 describes the system under study, the operational premises, the qualitative BowTie construction, and the quantitative-coupling procedure. Section 4 reports the populated BowTie diagram and the values of the quantitative indicators for the AP 1.8 case. Section 5 discusses the implications for cylinder-head design, compares the proposed approach with FMEA/RPN, and states the limitations of the present analytical study. Section 6 closes the paper. The work extends the case-study tradition of *Engineering Failure Analysis* into prospective design-stage barrier evaluation, complementary to the post-failure analyses of Vélez Godiño et al. (2019); Soffritti et al. (2018); Xing et al. (2021); Zhao et al. (2023) that informed the present formulation.

2. Background

2.1. Valve-train failure modes at high rotational speed

At elevated rotational speeds, the valve train of an SI engine moves from a quasi-static loading regime, in which the valve simply tracks the cam profile, into an inertia-dominated regime in which the spring force is the only mechanism that keeps the follower in contact with the cam (Heywood, 1988; Taylor, 1985). The classical failure modes that emerge in this regime are the following.

Valve float When the spring force is no longer sufficient to overcome the inertial force of the moving mass on the deceleration ramp of the cam, the follower separates from the cam. The valve then moves freely under the combined effect of the spring force and the residual inertia, and re-contact with the cam occurs in an uncontrolled manner. Float reduces the effective valve-opening profile and introduces large impacts at re-contact.

Valve bounce Bounce is the impact-driven oscillation of the valve at seating. At high rpm the closing velocity grows with the cam acceleration, and the combination of spring stiffness and seat compliance can result in multiple short re-openings before the valve finally seats. Bounce degrades sealing, accelerates seat wear, and disturbs the gas-exchange process.

Coil bind Coil bind is the condition in which adjacent coils of the valve spring are in metal-to-metal contact. When this happens, the spring stops behaving as a linear element: stiffness rises sharply and the dynamic load is transferred directly through the coil stack rather than through the spring deformation. Even an instantaneous approach to coil bind can produce a load spike sufficient to damage the spring, the retainer, or the cam lobe.

Spring fatigue The valve spring operates under fluctuating shear loading at the cam frequency, oscillating between a non-zero seat-preload stress and the full-lift stress; the alternating component is therefore superimposed on a positive mean stress and is more demanding than a fully-reversed loading of the same amplitude. At 10 000 rpm a single spring undergoes of the order of 10^8 cycles in 200 hours of operation. Fatigue resistance therefore governs the long-term life, and any local stress concentration (surface defect, shot-peening discontinuity, corrosion) can dominate the failure process (Budynas and Nisbett, 2020).

Resonance with cam harmonics The cam profile contains harmonics at integer multiples of the camshaft rotational frequency. If a harmonic coincides with the natural frequency of the spring, the spring enters resonance, the dynamic deflection grows beyond the design displacement, and coil bind and surge become likely (Heywood, 1988).

These five modes are not independent: float can trigger an impact that exceeds the residual coil-bind clearance, and resonance can amplify both. The BowTie methodology, introduced next, provides a unified language to describe how each mode contributes to a single top event.

2.2. The BowTie methodology

The BowTie methodology organises a risk analysis around a single top event. Threats (causes) are arranged on the left side of the bow, connected to the top event through preventive barriers; consequences are arranged on the right side, connected to the top event through mitigative barriers. The diagram resembles a bowtie, hence the name. The methodology has its origins in process-safety analysis and was consolidated as a general risk-management framework by the review of de Ruijter and Guldenmund (2016); it has since been applied to mechanical-reliability problems where the chain of causation deserves explicit visualisation (Aust and Pons, 2019).

Compared with FMEA, BowTie offers three advantages relevant to the present problem:

1. *Causal chain*: BowTie distinguishes threats, preventive barriers, consequences, and mitigative barriers as separate elements; FMEA collapses this into a single severity–occurrence–detectability triplet.
2. *Barrier-centric view*: each preventive barrier can be associated with a measurable indicator and a pass criterion, supporting the quantitative coupling proposed in this work.
3. *Communication*: the diagrammatic representation is accessible to designers, operators, and reliability engineers simultaneously, which is valuable in the context of a master's research that aims to bridge mechanical design and reliability analysis.

The classical BowTie is qualitative. To compensate for that limitation, several quantitative extensions have been proposed; in particular, Khakzad et al. (2013) map the BowTie into a Bayesian network so that the diagram becomes dynamic and probabilistic, and the review of de Ruijter and Guldenmund (2016) catalogues the alternative extensions ranging from layer-of-protection analysis (LOPA) to fault-tree-based barrier reliability. The approach adopted here is more pragmatic: each preventive barrier of the diagram is mapped to a deterministic indicator computed from the analytical sizing of the component, with a literature-anchored pass criterion. This avoids the need for probabilistic data that is not available for the bespoke high-performance cylinder head under study.

2.3. Related work

Several authors have addressed valve-train and engine-component reliability from complementary perspectives. The work of Vélez Godiño et al. (2019) on overhead valve trains in urban buses is a representative case study published in *Engineering Failure Analysis*: the authors use field data and metallurgical inspection to attribute the recurrent failure of an in-service valve train to a combination of operational misuse and inadequate inspection intervals, but they stop short of organising the findings around a barrier-based diagram. Soffritti et al. (2018) report a four-cylinder diesel engine in which excessive wear at the rocker-arm/pushrod and rocker-arm/valve interfaces appeared after only 1 000 hours of operation; their root-cause analysis points to valve misalignment with respect to the seat insert as the dominant trigger. Both studies confirm that valve-train failures, once initiated, propagate quickly into other components and motivate the need for a preventive analysis framework.

Two further studies in *Engineering Failure Analysis* address elements of the valve-train system that are central to the present work. Xing et al. (2021) investigate environment-assisted cracking in high-strength valve springs and show that the fatigue limit adopted at the design stage must be discounted to account for surface condition and lubricant chemistry; this finding directly supports the fatigue factor of safety adopted in Section 3.4. Zhao et al. (2023) report lubrication failure in the camshaft bearings of a V6 diesel engine, identifying an asperity-contact regime that violates the design assumption of full hydrodynamic lubrication and reduces the effective service life of the bearing pair; their result reinforces the importance of the lubrication-related preventive barriers shown on the left side of the BowTie diagram of Section 3.3. Ko et al. (2022) extend the spring-fatigue discussion beyond *Engineering Failure Analysis*, showing in *Scientific Reports* that surface flaws as small as a few micrometres on the oil-tempered wire of an automotive valve spring can reduce the fatigue life by an order of magnitude; this result motivates the conservative factor of safety adopted in Table 2.

The methodological literature on the BowTie diagram itself has matured in parallel. The review by de Ruijter and Guldenmund (2016) remains the canonical reference for the qualitative formulation and catalogues the variations adopted in the chemical, aerospace, and nuclear industries. Khakzad et al. (2013) propose a mapping of the BowTie into a Bayesian network so that the static qualitative diagram becomes dynamic and probabilistic; their construction is adopted in part by the BowTie tool used in the offshore industry. Aust and Pons (2019) apply the methodology to aircraft engine borescope inspection and show that, even in the absence of probabilistic data, the BowTie diagram is an effective communication artefact between maintenance planners, inspectors, and reliability engineers.

It is also relevant to acknowledge a parallel branch of the literature that extends the conventional FMEA in different directions: fuzzy-FMEA formulations (Xing et al., 2021) and analytic-hierarchy-process-based weighting (AHP-FMEA) have been used in mechanical reliability to mitigate the loss of resolution of the multiplicative Risk Priority Number (RPN). The present work is complementary to those extensions: it does not aim to refine the FMEA score but to replace the single-score logic with a barrier-by-barrier deterministic check, and is therefore positioned closer to the BowTie tradition than to fuzzy/AHP-FMEA.

To the best of the authors' knowledge, no published study applies the BowTie methodology to the valve train of a high-rpm spark-ignition cylinder head, and no study couples the BowTie diagram to the analytical sizing of the intake valve and the valve spring as a reproducible barrier-evaluation procedure. The present work fills both gaps.

3. Methodology

3.1. System under study

The system under study is the valve train of a high-performance cylinder head conceived for the Volkswagen AP 1.8 platform. The base engine has a bore of 81.0 mm, a stroke of 86.4 mm, a total displacement of 1 781 cm^3, and a four-cylinder in-line architecture (Porto, 2026, § 1.1). The cylinder head is of the pent-roof type, with inclined valves, asymmetric quench, gasoline direct injection (GDI), and an iridium spark plug; these features establish the mechanical and thermal context of the valve train but are not the subject of the risk analysis presented here. The reader is referred to Porto (2026) for the full description of the cylinder head and the air-flow optimisation.

3.2. Mechanical and operational premises

The premises adopted for the present risk analysis are summarised in Table 1 and follow the design choices of the dissertation (Porto, 2026) and the associated sizing notes for the intake valve and the valve springs.

3.3. Construction of the qualitative BowTie

The BowTie diagram is constructed around the top event *loss of follower contact between valve and cam*. This formulation intentionally aggregates the three failure modes of Section 2.1 that share a single mechanical signature: valve float, valve bounce, and approach to coil bind. The remaining modes (resonance and fatigue) act as enabling threats rather than as the top event itself.

The threats placed on the left side of the bow are derived from the classical literature (Heywood, 1988; Taylor, 1985; Heisler, 2002) and from the sizing constraints of the AP 1.8 head. They are: (i) insufficient spring stiffness for the inertial regime; (ii) excessive moving mass; (iii) aggressive cam profile; (iv) spring heating in operation; (v) out-of-tolerance thermal clearance; and (vi) resonance between cam harmonics and spring natural frequency. The preventive barriers placed between threats and the top event are: (i) dynamic sizing of the spring relative to the cam harmonics; (ii) dual or nested springs; (iii) low-mass tappets; (iv) high-fatigue-resistance spring material; and (v) modal analysis and dynamometer validation.

The consequences placed on the right side of the bow are: (i) valve–piston collision; (ii) catastrophic cylinder-head failure; (iii) immediate loss of compression; (iv) oil contamination by debris; and (v) unscheduled engine shutdown. The mitigative barriers between the top event and these consequences are: (i) electronic rev limiter; (ii) cam-position sensor; (iii) operating envelope defined at design stage; (iv) torque or fuel cut-off; and (v) predictive maintenance via vibration signals.

3.4. Quantitative coupling layer

To overcome the qualitative limitation of the classical BowTie, each of the five preventive barriers is associated with a deterministic indicator and a pass criterion, summarised in Table 2. The indicators are computed from the analytical sizing of the intake valve and of the valve springs, which is reported in the underlying dissertation (Porto, 2026) and in two associated sizing notes on the intake valve and the spring.

The dynamic-ratio criterion $R \geq 1.20$ corresponds to the project specification adopted in the AP 1.8 sizing and is consistent with the design practice for high-speed valve trains (Heywood, 1988; Taylor, 1985; Heisler, 2002).

The natural-frequency criterion follows the rule of thumb that the first surge mode of the spring should sit above the dominant cam harmonics (Heywood, 1988). Two equivalent formulations are available in the literature: the one-degree-of-freedom approximation $f_n = \frac{1}{2}\sqrt{k/m_s}$ for a coil spring with both ends fixed (Budynas and Nisbett, 2020, § 10-26), and the continuous-medium expression $f_n = (d/(2\pi N D_m^2))\sqrt{G/(2\rho)}$ (Heywood, 1988, § 6.6), which reduce algebraically to the same result when m_s in the first form is taken as the active-coil mass. In the present analysis the conservative form $f_n = \frac{1}{2}\sqrt{k/m_{s,\text{total}}}$ is adopted with the *total* spring mass (including end coils), which yields the lower bound of the surge frequency and is the more demanding test for the rule-of-thumb threshold.

The coil-bind margin of 1.5–2.0 mm is the practical specification adopted in the AP 1.8 sizing to absorb manufacturing tolerances, mounting variations, and dynamic excitations typical of operation at 10 000 rpm. The fatigue criterion uses the Goodman alternating-mean factor of safety (Eq. 5) with the shear endurance limit and the ultimate shear strength of ASTM A877/A877M chrome–silicon valve-spring steel after shot peening and preset (Budynas and Nisbett, 2020; Xing et al., 2021; Ko et al., 2022). This paper does not

Table 1
Mechanical and operational premises adopted for the risk analysis of the AP 1.8 valve train. Values taken from the dissertation (Porto, 2026) and the associated sizing notes for the intake valve and the valve springs.

Engine and operating regime		*Dual valve-spring assembly (Cr–Si steel, ASTM A877/A877M)*	
Bore	81.0 mm	Outer spring stiffness k_{out}	38.06 N mm^{-1}
Stroke	86.4 mm	Inner spring stiffness k_{in}	61.43 N mm^{-1}
Total displacement	1 781 cm^3	Combined stiffness k_{total}	99.49 N mm^{-1}
Cylinders	4 in line	Seat preload F_{seat}	780 N
Maximum engine speed	10 000 rpm	Force at full lift F_{open}	1 985.8 N
Camshaft fund. freq. $f_{\text{cam,max}}$	83.3 Hz	Outer spring solid height	33.6 mm
Compression ratio	14.5 : 1	Inner spring solid height	32.0 mm
Valves, rocker and cam		Outer spring height at full lift	34.38 mm
Intake valve material	Ti–6Al–4V	Total spring assembly mass	152.72 g
Exhaust valve material	INCONEL 751	Effective spring mass ($m_{\text{molas}}/3$)	50.91 g
Roller rocker / retainer / keeper	7075–T6 aluminium		
Camshaft material	SAE 8620 case-hardened		
Maximum valve lift	12.12 mm	*Direct translational mass (intake side)*	
Cam accel. peak (positive)	14 544 m s^{-2}	Valve + retainer + keeper	43.48 g
Cam decel. peak (critical)	20 362 m s^{-2}	With reflected rocker	49.59 g
		Direct translational mass (exhaust side)	
		Valve + retainer + keeper	54.25 g
		With reflected rocker	60.36 g

Table 2
Quantitative indicators associated with the preventive barriers of the BowTie diagram. The pass criteria follow conventional design rules in the internal-combustion-engine literature and the project specification of the AP 1.8 sizing.

Barrier	Indicator	Pass criterion
Dynamic follower contact	$R = F_{\text{spring,crit}}/F_{\text{inertia,crit}}$	$R \geq 1.20$ (project)
Spring natural frequency	$f_n = \frac{1}{2}\sqrt{k/m_s}$	$f_n/f_{\text{cam,max}} \geq 4$ (Heywood, 1988)
Coil-bind margin (outer)	$M = H_{\text{open}} - H_{\text{cb}}$	$M \geq 1.5$ mm at 10 000 rpm
Fatigue (Goodman)	$1/n_f = \tau_a/S_{se} + \tau_m/S_{su}$	$n_f \geq 1$ (Budynas and Nisbett, 2020)
Hot clearance†	$c(T_{\max})$	$c(T_{\max}) \geq 0$ over the full map

†Hot clearance is treated in this paper as a *qualitatively verified* barrier outside the deterministic layer; only a first-order order-of-magnitude estimate is provided in Section 4.2. A complete differential thermal model is identified as a priority direction for future work in Section 5.3.

propose new criteria; it organises them around the BowTie diagram.

3.5. Cross-check procedure

For the AP 1.8 head, the values of all indicators in Table 2 are computed under the operating envelope of Table 1. A barrier is declared *active* if the indicator satisfies the pass criterion; otherwise it is declared *at risk*. The single barrier with the smallest absolute margin is declared the *critical barrier*: it is the first that would fail under a parametric drift of the inputs (mass increase, stiffness loss through heating, harder cam profile, etc.). A tornado plot of the sensitivity of each indicator to its dominant input is reported in Section 4 (Fig. 2) to make the identification of the critical barrier transparent.

The qualitative BowTie of Section 3.3 and the quantitative layer of Section 3.4 are then cross-checked against a conventional FMEA/RPN ranking of the same threats. The comparison is discussed in Section 5.2.

4. Results

4.1. Populated BowTie diagram

Figure 1 shows the populated BowTie diagram for the AP 1.8 valve train at 10 000 rpm, around the top event *loss of follower contact between valve and cam*. The threats, preventive barriers, consequences, and mitigative barriers are those of Section 3.3.

4.2. Quantitative evaluation of the preventive barriers

The five quantitative indicators of Table 2 were evaluated for the AP 1.8 head under the operating envelope of Table 1. The numerical results are reported in Table 3 and discussed below.

Dynamic follower contact. At the critical deceleration point of the cam ($a_{\max,-} = 20\,362$ m s^{-2}, $L_- = 8.747$ mm),

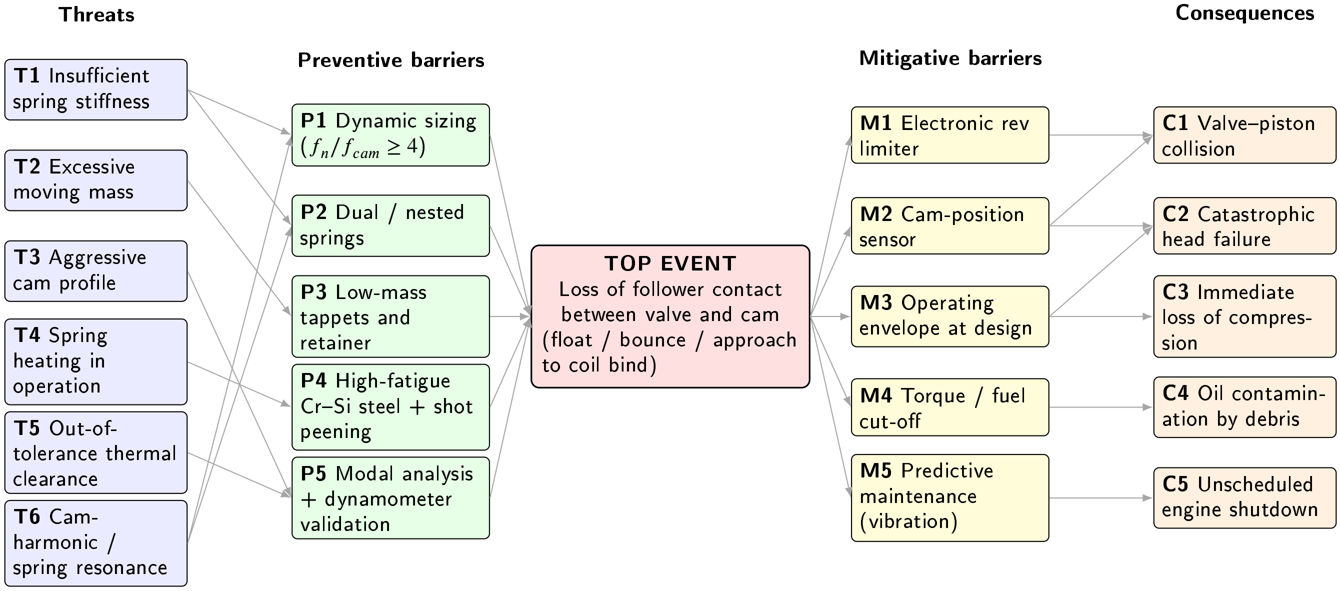

Figure 1: BowTie diagram of the valve train of the AP 1.8 cylinder head operating at 10 000 rpm. Threats (T1–T6) and preventive barriers (P1–P5) are arranged on the left side; consequences (C1–C5) and mitigative barriers (M1–M5) are arranged on the right side. The top event is the loss of follower contact between the valve and the cam, which encompasses valve float, valve bounce, and approach to coil bind.

the spring force is

$$F_{\text{spring},-} = F_{\text{seat}} + k_{\text{total}}\, L_{-} = 780 + 99.49 \times 8.747 = 1\,650.2\,\text{N}, \tag{1}$$

and the inertial force on the intake side, computed with the direct translational mass of 43.48 g, is $F_{i,d,-,\text{adm}} = 885.3$ N, giving a dynamic ratio of $R_{d,-,\text{adm}} = 1.864$. With the equivalent mass that includes the reflected rocker (49.59 g), the dynamic ratio decreases to $R_{\text{eq},-,\text{adm}} = 1.634$. On the exhaust side, the larger mass (54.25 g translational direct) yields $R_{d,-,\text{esc}} = 1.494$ and $R_{\text{eq},-,\text{esc}} = 1.343$. All four values are above the $R \geq 1.20$ project criterion, so the follower-contact barrier is satisfied with a margin of at least 12 % (in the worst case, the exhaust side with reflected rocker).

Spring natural frequency. The fundamental surge mode of a helical compression spring with both ends fixed is the first axial standing wave on the helical wire. For a spring of stiffness k and active mass m_s, the wave equation yields a fundamental frequency $\omega_1 = \pi\sqrt{k/m_s}$ rad s^{-1}, hence $f_n = \omega_1/(2\pi) = \frac{1}{2}\sqrt{k/m_s}$ Hz, which is the form given in Budynas and Nisbett (2020, Eq. 10-26). This expression is algebraically equivalent to the continuous-medium derivation $f_n = (d/(2\pi N D_m^2))\sqrt{G/(2\rho)}$ of Heywood (1988, § 6.6): substituting $k = Gd^4/(8D_m^3 N)$ and the active-coil wire mass $m_s = (\pi^2/4)\, N D_m d^2 \rho$ in $f_n = \frac{1}{2}\sqrt{k/m_s}$ recovers Heywood's expression exactly. In the present analysis the conservative lower bound is adopted by using the *total* spring mass (94.34 g for the outer and 47.11 g for the inner spring), which includes the inactive end turns and yields the lower bound of the surge frequency:

$$f_{n,\text{out}} = \frac{1}{2}\sqrt{38\,060\,\text{N}\,\text{m}^{-1}/0.09434\,\text{kg}} \approx 318\,\text{Hz}, \tag{2}$$

$$f_{n,\text{in}} = \frac{1}{2}\sqrt{61\,430\,\text{N}\,\text{m}^{-1}/0.04711\,\text{kg}} \approx 571\,\text{Hz}. \tag{3}$$

For comparison, replacing $m_{s,\text{total}}$ with the conventional effective mass $m_s/3$ (Rayleigh's quotient for a uniform coil spring with one end fixed) would yield $f_{n,\text{out}} \approx 551$ Hz and $f_{n,\text{in}} \approx 988$ Hz. Adopting the total spring mass therefore penalises the rule-of-thumb test against the cam harmonics by a factor of $\sqrt{3} \approx 1.73$ and is treated here as an explicit robustness check rather than a tolerance estimate. The camshaft fundamental frequency at 10 000 rpm engine speed ($f_{\text{cam,max}} = 10\,000/(2 \times 60) = 83.3$ Hz, the factor two accounting for the 4-stroke cam-to-engine speed ratio) gives the natural-frequency margins $f_{n,\text{out}}/f_{\text{cam,max}} \approx 3.81$ and $f_{n,\text{in}}/f_{\text{cam,max}} \approx 6.85$. The outer spring sits just below the conventional $f_n/f_{\text{cam}} \geq 4$ threshold (Heywood, 1988) and is therefore at the limit of the preventive barrier; the inner spring is comfortably above the threshold against the cam fundamental. However, real cam profiles contain harmonic content up to the sixth to eighth harmonic, and the proximity of $f_{n,\text{in}} \approx 571$ Hz to the sixth harmonic ($6\, f_{\text{cam,max}} = 500$ Hz; about 14 % separation) is discussed in Section 4.4 and noted as a residual concern.

Coil-bind margin. At the maximum valve lift of 12.12 mm, the outer spring is compressed to a height of $H_{\text{open}} =$

34.38 mm. With a solid height of $H_{\mathrm{cb,out}} \approx 33.6$ mm (from $N_{t,\mathrm{out}}\, d_{\mathrm{out}}$), the residual margin to coil bind is

$$M_{\mathrm{out}} = H_{\mathrm{open}} - H_{\mathrm{cb,out}} = 34.38 - 33.6 = 0.78 \text{ mm}, \quad (4)$$

which is below the project specification of $M \geq 1.5$ mm at 10 000 rpm. In the concentric dual-spring assembly the inner and outer springs share the same retainer and therefore undergo the same axial compression at full lift, so H_{open} is identical for both. The inner spring, with $H_{\mathrm{cb,in}} \approx 32.0$ mm, therefore yields $M_{\mathrm{in}} = H_{\mathrm{open}} - H_{\mathrm{cb,in}} = 34.38 - 32.0 = 2.38$ mm, comfortably above the criterion. The outer-spring coil-bind margin is therefore the only barrier that does not satisfy its pass criterion under the current configuration of the AP 1.8 head.

Fatigue stress. The peak shear stress at full lift, computed with the Wahl-corrected formula $\tau_{\max} = K_w\, 8FD_m/(\pi d^3)$, is $\tau_{\mathrm{out}} \approx 655$ MPa for the outer spring ($K_{w,\mathrm{out}} = 1.239$, $D_{m,\mathrm{out}} = 30.2$ mm, $d_{\mathrm{out}} = 4.8$ mm) and $\tau_{\mathrm{in}} \approx 1\,232$ MPa for the inner spring ($K_{w,\mathrm{in}} = 1.329$, $D_{m,\mathrm{in}} = 19.0$ mm, $d_{\mathrm{in}} = 4.0$ mm). Because the spring operates between a non-zero seat preload and the full-lift load, the proper criterion is the alternating-mean Goodman criterion (Budynas and Nisbett, 2020, § 10-7) rather than a single-point comparison with an allowable. With the corresponding seat-load shear stresses $\tau_{\mathrm{out,seat}} \approx 257$ MPa and $\tau_{\mathrm{in,seat}} \approx 484$ MPa, the alternating and mean components are $\tau_{a,\mathrm{out}} = 199$ MPa, $\tau_{m,\mathrm{out}} = 456$ MPa and $\tau_{a,\mathrm{in}} = 374$ MPa, $\tau_{m,\mathrm{in}} = 858$ MPa.

For shot-peened ASTM A877/A877M chrome–silicon valve-spring steel, typical strengths are an ultimate tensile strength $S_{ut} \approx 1\,900$ MPa, an ultimate shear strength $S_{su} \approx 0.67\, S_{ut} \approx 1\,270$ MPa, and a shear endurance limit after shot peening and surface control of $S_{se} \approx 465$ MPa (Budynas and Nisbett, 2020, § 10-30). The Goodman factor of safety is then

$$\frac{1}{n_f} = \frac{\tau_a}{S_{se}} + \frac{\tau_m}{S_{su}}. \quad (5)$$

For the outer spring this gives $1/n_{f,\mathrm{out}} = 199/465 + 456/1\,270 \approx 0.428 + 0.359 = 0.787$, hence $n_{f,\mathrm{out}} \approx 1.27$ (active, with positive but not generous margin). For the inner spring, $1/n_{f,\mathrm{in}} = 374/465 + 858/1\,270 \approx 0.804 + 0.676 = 1.480$, hence $n_{f,\mathrm{in}} \approx 0.68$, which is below unity and indicates that the baseline configuration of the inner spring does not satisfy the Goodman criterion. The inner-spring fatigue barrier is therefore reclassified as **at risk** in Table 3, alongside the outer-spring coil-bind margin. This finding is consistent with Xing et al. (2021); Ko et al. (2022), who report that small surface flaws and environment-assisted cracking substantially reduce the fatigue life of high-strength valve springs loaded close to their nominal allowable.

Hot clearance. A complete thermal expansion model of the cylinder head and valve train is outside the scope of the present study, but a first-order estimate of the magnitude can be obtained from the linear thermal expansion of the longest element of the train, the exhaust valve stem (length $L_h = 180$ mm, INCONEL 751 with $\alpha \approx 13 \times 10^{-6}$ K^{-1}). For an exhaust-side temperature rise of $\Delta T \approx 600$ K above the cold-assembly condition, the stem elongates by $\Delta L = \alpha L_h \Delta T \approx 1.4$ mm. The corresponding cold mechanical clearance must therefore be at least $\sim$1.5 mm to maintain a positive hot clearance, a value compatible with the design clearance typically specified for high-rpm performance heads (Heisler, 2002). This worst-case estimate neglects the thermal expansion of the aluminium cylinder head, which acts in the opposite sense and partially compensates the valve-stem elongation; including it would relax the cold-clearance requirement. A dedicated differential thermal-mechanical analysis is left for future work and listed in the Limitations of Section 5.3.

4.3. Critical-barrier identification

A tornado plot of the sensitivity of each indicator to its dominant input is shown in Fig. 2. For each barrier, the input perturbed by $\pm 10\,\%$ is the dominant parameter governing the indicator: the spring stiffness k for the natural-frequency margin, the maximum cam lift $L_{\mathrm{cam}}^{\max}$ for the coil-bind margin, the moving mass m for the dynamic-ratio margin, and the shear-stress amplitude τ_a for the Goodman fatigue margin. The $\pm 10\,\%$ perturbation is adopted as a uniform stress test rather than a tolerance estimate, providing an upper-bound sensitivity envelope for the worst-case parameter drift. The actual ASTM A877/A877M specification tolerances are tighter and unequal by parameter (typically $\pm 1.5\,\%$ on free length, $\pm 0.5\,\%$ on wire diameter, and $\pm 3\,\%$ on shear modulus); a refined sensitivity analysis based on those non-uniform tolerances is identified as future work. The bar length corresponds to the absolute change in the residual margin under that perturbation. The analysis identifies **three** barriers as exposed to parametric drift: the *outer-spring coil-bind margin* ($M_{\mathrm{out}} = 0.78$ mm, violating its pass criterion already in the baseline configuration), the *inner-spring fatigue Goodman factor of safety* ($n_{f,\mathrm{in}} = 0.68$, below unity in the baseline configuration), and the *outer-spring natural-frequency margin* ($f_{n,\mathrm{out}}/f_{\mathrm{cam,max}} = 3.81$, just below the conventional 4× rule of thumb).

The findings indicate that the AP 1.8 dual-spring assembly has *two* concurrent at-risk barriers in the baseline configuration: the outer-spring coil-bind margin and the inner-spring fatigue factor of safety. They are not redundant: they affect different springs (outer vs inner) and act through different physical mechanisms (geometric clearance vs cyclic shear loading). The outer-spring natural-frequency margin sits just below the conventional threshold and is treated as a residual concern. The sizing note for the springs already prescribes the geometric correction for the outer-spring coil-bind — raising the installed height to $H_{\mathrm{inst,ideal}} = 47.72$ mm and the free length to $L_{0,\mathrm{ideal}} = 55.56$ mm to recover an $M \approx 2.0$ mm margin. The inner-spring fatigue concern is not addressed by that geometric change and would require a different intervention: increasing the wire diameter d_{in}, reducing the spring index $C_{\mathrm{in}} = D_{m,\mathrm{in}}/d_{\mathrm{in}}$, or adopting a higher-grade chrome–silicon–vanadium steel with a larger

Table 3
Quantitative results for the five preventive barriers of the BowTie diagram applied to the AP 1.8 valve train at 10 000 rpm. Values traced to the sizing of the intake valve and the valve springs.

Barrier	Indicator	Value	Status
Dynamic follower contact (intake, direct mass)	$R_{d,-,\mathrm{adm}}$	1.864	active (≥ 1.20)
Dynamic follower contact (intake, with rocker)	$R_{\mathrm{eq},-,\mathrm{adm}}$	1.634	active (≥ 1.20)
Dynamic follower contact (exhaust, direct mass)	$R_{d,-,\mathrm{esc}}$	1.494	active (≥ 1.20)
Dynamic follower contact (exhaust, with rocker)	$R_{\mathrm{eq},-,\mathrm{esc}}$	1.343	active (≥ 1.20)
Natural frequency (outer spring)	$f_{n,\mathrm{out}}/f_{\mathrm{cam,max}}$	3.81	at limit (< 4)
Natural frequency (inner spring)	$f_{n,\mathrm{in}}/f_{\mathrm{cam,max}}$	6.85	active (≥ 4)
Coil-bind margin (outer spring)	M_{out}	0.78 mm	**at risk** (< 1.5 mm)
Coil-bind margin (inner spring)	M_{in}	2.38 mm	active (≥ 1.5 mm)
Fatigue Goodman FoS (outer spring)	$n_{f,\mathrm{out}}$	1.27	active (≥ 1)
Fatigue Goodman FoS (inner spring)	$n_{f,\mathrm{in}}$	0.68	**at risk** (< 1)
Hot clearance†	$c(T_{\mathrm{max}})$	—	qualitatively verified (see Sec. 4.2)

†Outside the deterministic layer; only a first-order order-of-magnitude estimate is reported.

shear endurance limit. The two findings together reinforce the value of the quantitative-coupled BowTie approach: a single-point comparison of τ_{max} with an upper-bound allowable, in the spirit of conventional FMEA, would have classified the inner-spring fatigue as nominally acceptable, whereas the Goodman criterion exposes the deficit unambiguously.

4.4. Lift profile and natural-frequency map

For completeness, Fig. 3 shows the position of the two critical points of the cam profile (positive-acceleration peak and deceleration peak) on the lift–versus–crank-angle plane; Fig. 4 compares the spring natural frequencies with the first cam harmonics across the rotational-speed range up to 10 000 rpm engine speed.

5. Discussion

5.1. Implications for cylinder-head design

The quantitative-coupled BowTie of Section 4 provides a single artefact that links each design decision in the high-performance cylinder head to the risk it controls. The analysis identified two concurrent at-risk barriers under the baseline configuration of the AP 1.8 head: (i) the *outer-spring coil-bind margin* ($M_{\mathrm{out}} = 0.78$ mm against a project criterion of $M \geq 1.5$ mm), and (ii) the *inner-spring fatigue Goodman factor of safety* ($n_{f,\mathrm{in}} \approx 0.68$ against $n_f \geq 1$). The outer-spring natural-frequency margin ($f_{n,\mathrm{out}}/f_{\mathrm{cam,max}} \approx 3.81$) sits just below the conventional rule-of-thumb threshold of 4 and is treated as a residual concern. The remaining barriers — dynamic follower contact ($R \geq 1.343$ in the worst case, against the project criterion of $R \geq 1.20$), inner-spring coil-bind margin ($M_{\mathrm{in}} = 2.38$ mm), and outer-spring fatigue ($n_{f,\mathrm{out}} \approx 1.27$) — satisfy their pass criteria with non-uniform margins.

The fact that the two at-risk barriers affect *different* springs (outer for coil-bind, inner for fatigue) and act through *different* physical mechanisms (geometric clearance vs cyclic shear loading) is significant: a design iteration that addresses only one of the two findings will leave the other unresolved. The sizing note for the AP 1.8 springs already prescribes the geometric correction for the outer-spring coil-bind (raising the installed height to $H_{\mathrm{inst,ideal}} = 47.72$ mm and the free length to $L_{0,\mathrm{ideal}} = 55.56$ mm to achieve $M \approx 2.0$ mm). The inner-spring Goodman deficit, by contrast, calls for a metallurgical or geometric change of a different nature: either increasing the wire diameter d_{in} to lower the shear stress at a given load, reducing the spring index $C_{\mathrm{in}} = D_{m,\mathrm{in}}/d_{\mathrm{in}}$, or specifying a higher-grade steel (e.g. chrome–silicon–vanadium ASTM A878) with a larger corrected shear endurance limit S_{se}. Concerning the residual outer-spring natural-frequency concern, a parallel correction would adjust either the active coil count $N_{a,\mathrm{out}}$ or the spring index C_{out} to push $f_{n,\mathrm{out}}$ further from the dominant cam harmonics. The BowTie diagram makes the three trade-offs simultaneously visible to the design team.

The remaining barriers retain non-zero margins and therefore satisfy the literature-anchored pass criteria, but Table 3 shows that the margins are not uniform. This non-uniformity is a useful input for design iteration: barriers with comfortable margins (e.g. the inner-spring coil-bind margin or the outer-spring fatigue stress) are candidates for relaxation if doing so frees mass or simplifies manufacturing; barriers close to the pass threshold should be the focus of dynamometer validation as soon as a physical prototype is available.

5.2. Comparison with traditional FMEA/RPN

Table 4 contrasts the present BowTie analysis with a conventional FMEA of the same six threats. Three observations follow.

1. *Granularity of the causal chain*: the BowTie diagram makes the linkage between threat, preventive barrier, top event, and consequence visible in a single artefact, whereas FMEA collapses the causal chain into the multiplicative RPN. For a high-rpm valve train, this granularity is necessary because several threats share the same preventive barriers (for example, both

Table 4
Comparison between the BowTie analysis presented in this work and a conventional FMEA of the same threats.

Aspect	FMEA / RPN	Quantitative BowTie
Causal chain	Implicit	Explicit
Severity assessment	Ordinal (1–10)	Continuous (deterministic)
Pass criterion	RPN threshold	Indicator-by-indicator
Reproducibility	Limited	Direct from sizing
Visualisation	Tabular	Diagrammatic
Identifies critical barrier	No	Yes (via tornado plot)

insufficient stiffness and *spring heating* are addressed by the spring-natural-frequency barrier).

2. *Quantitative anchoring*: in FMEA, severity, occurrence, and detectability are typically assigned on integer scales (1–10) by expert judgement. The quantitative-coupled BowTie replaces these ordinal scores with deterministic indicators that can be reproduced by any reader who has access to the sizing calculations.
3. *Communication*: the BowTie diagram is accessible to designers, operators, and reliability engineers without prior familiarity with the RPN convention. This is a practical advantage in the academic context of a master's research that aims to articulate mechanical design and reliability analysis.

A concrete numerical illustration of why the conventional RPN collapses information that the BowTie preserves is given in Table 5. The six threats T_1–T_6 of the BowTie diagram are scored on the conventional 1–10 ordinal scales of severity (S), occurrence (O), and detectability (D) following the engineering judgement reasoning that (i) all six threats lead to the same top event with catastrophic consequences, hence S is uniformly high (8–10); (ii) O varies with how sensitive each threat is to parametric drift; and (iii) D is uniformly low (3–4) because the failure modes are mechanical and predominantly silent before the event of failure. The resulting RPNs fall in a narrow band (96–240), which makes the prioritisation ambiguous: a triaging team using a conventional 100–125 RPN threshold would not be able to distinguish reliably between threats addressed by the outer-spring coil-bind barrier (T_1, RPN = 144) and threats addressed by the inner-spring fatigue barrier (e.g. T_4, RPN = 150), even though the quantitative-coupled BowTie identifies both as concurrently at risk through different physical mechanisms. The RPN ranking, in other words, masks the very finding that the BowTie analysis surfaces.

The two methods are not exclusive: FMEA remains useful for triaging a large number of failure modes when probabilistic data is unavailable, and BowTie is most valuable when a small number of high-severity events deserves a detailed barrier analysis. For the high-performance cylinder head studied here, the BowTie approach is preferred because the failure modes converge on a single top event whose consequences are catastrophic.

Table 5
Conventional FMEA scoring (1–10 ordinal scales) of the six threats T_1–T_6 of the BowTie diagram. The Risk Priority Numbers fall in a narrow band that does not reliably discriminate between threats addressed by the two at-risk barriers identified in Section 4. Severity, occurrence, and detectability scores reflect engineering judgement and are illustrative; they are not the result of a formal expert panel.

Threat	S	O	D	RPN
T_1 Insufficient spring stiffness	9	4	4	144
T_2 Excessive moving mass	9	3	4	108
T_3 Aggressive cam profile	10	3	4	120
T_4 Spring heating in operation	10	5	3	150
T_5 Out-of-tolerance thermal clearance	8	4	3	96
T_6 Cam-harmonic / spring resonance	10	6	4	240

A parallel branch of the literature has extended the conventional FMEA through fuzzy-set formulations and AHP-based weighting to address the loss of resolution of the multiplicative RPN (Xing et al., 2021). Compared with these approaches, the proposed quantitative-coupled BowTie does not refine the FMEA score; it *replaces* the single-score logic with a barrier-by-barrier deterministic check that is reproducible from the sizing calculations. The two routes are therefore complementary: fuzzy/AHP-FMEA improves the prioritisation among many failure modes, while the present BowTie improves the auditability of the design margins around a single critical event.

5.3. Limitations

The present analysis has the following limitations, which should be considered when interpreting the results.

- *No experimental validation.* The quantitative indicators are computed from the analytical sizing of the valve and the spring; no dynamometer or rig testing of the high-performance cylinder head has been performed. Validation under real operating conditions is left for future work.
- *Single operating point.* The pass criteria are evaluated at the maximum design speed of 10 000 rpm. Off-design behaviour (part-load, cold start, overspeed) is not considered.
- *Deterministic barrier indicators.* The barriers are treated as deterministic and the sensitivity analysis of Section 4.3 is reported as a uniform $\pm 10\,\%$ perturbation of each dominant input, adopted as a uniform stress test rather than a tolerance estimate; the actual ASTM A877/A877M specification tolerances are tighter and unequal by parameter (typically $\pm 1.5\,\%$ on free length, $\pm 0.5\,\%$ on wire diameter, and $\pm 3\,\%$ on shear modulus). A probabilistic Monte Carlo extension that propagates those non-uniform tolerance distributions and the operational variability (thermal load, oil condition) into a probability of failure for

each barrier would refine the analysis substantially and is identified as the priority direction of future work; the indicator forms in Table 2 are designed so that this extension can be applied without methodological rework.

- *Single design instance.* The analysis is conducted on a single bespoke cylinder head ($n = 1$). The methodology itself is transposable to other high-rpm SI heads, but the numerical findings (critical barrier identity, indicator margins) are case-specific and should not be generalised without re-evaluation.

- *Out of scope by design choice.* Knock and other abnormal combustion phenomena, low-speed pre-ignition, IoT-based monitoring, artificial-intelligence-supported diagnostics, and the multi-objective optimisation of the cylinder-head geometry are explicitly out of scope. The reader is referred to the underlying dissertation (Porto, 2026) for the air-flow and combustion analysis of the same head.

6. Conclusion

This work proposed a quantitative-coupled BowTie risk analysis of the valve train of a high-performance cylinder head conceived for the Volkswagen AP 1.8 engine operating at 10 000 rpm. The qualitative BowTie diagram organises six threats and five preventive barriers around the top event *loss of follower contact between valve and cam*, together with five consequences and five mitigative barriers. To overcome the qualitative limitation of the classical BowTie, each preventive barrier was associated with a deterministic indicator (dynamic follower-contact ratio, spring natural frequency, coil-bind margin, and Goodman fatigue factor of safety), with the hot clearance treated as a qualitatively verified barrier outside the deterministic layer, all anchored in literature-based pass criteria.

For the AP 1.8 head under the operating envelope adopted, the analysis identified two concurrent at-risk preventive barriers in the baseline configuration: (i) the *outer-spring coil-bind margin* ($M_{\text{out}} = 0.78$ mm against the project criterion $M \geq 1.5$ mm); and (ii) the *inner-spring fatigue Goodman factor of safety* ($n_{f,\text{in}} \approx 0.68$ against $n_f \geq 1$ for the fluctuating shear loading between seat preload and full-lift force). The two findings affect different physical components and require independent design interventions. The *outer-spring natural frequency* ($f_{n,\text{out}}/f_{\text{cam,max}} \approx 3.81$) emerges as a residual concern just below the conventional 4× rule of thumb. The remaining barriers — dynamic follower contact ($R \geq 1.343$ in the worst case), inner-spring coil-bind margin, and outer-spring fatigue ($n_{f,\text{out}} \approx 1.27$) — satisfy the pass criteria with non-uniform margins. The comparison with a conventional FMEA/RPN of the same threats showed that the quantitative-coupled BowTie offers (i) explicit visualisation of the causal chain, (ii) reproducibility from the sizing calculations, and (iii) identification of multiple concurrent critical barriers via a sensitivity analysis that a single-score RPN would have missed.

The proposed methodology is reproducible and can be transposed to other high-performance cylinder heads operating at high rotational speed, provided the analytical sizing of the valve train is available. Future work will address experimental validation of the critical barrier on a dynamometer, probabilistic extension of the barrier indicators to account for manufacturing and operational variability, and integration of the proposed risk analysis as a constraint in the multi-objective optimisation of the cylinder-head geometry.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

No new data were generated in this study. Analytical inputs are drawn from the sizing calculations referenced in the text.

Generative AI disclosure

During the preparation of this work the authors used Claude (Anthropic) to support the structuring of the manuscript outline and language editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Acknowledgments

The authors thank the Centre for Engineering and Computing of the Universidade Católica de Petrópolis and the Graduate Programme in Engineering Systems for supporting the underlying Master's research project from which the analytical sizing of the AP 1.8 cylinder head was derived.

CRediT authorship contribution statement

Darlan Costa Porto: Conceptualization, Methodology, Formal analysis, Investigation, Visualization, Writing – original draft. **Heitor Oliveira Gonçalves:** Methodology, Validation, Formal analysis, Writing – review & editing, Supervision. **José Cristiano Pereira:** Conceptualization, Methodology, Supervision, Writing – review & editing, Project administration.

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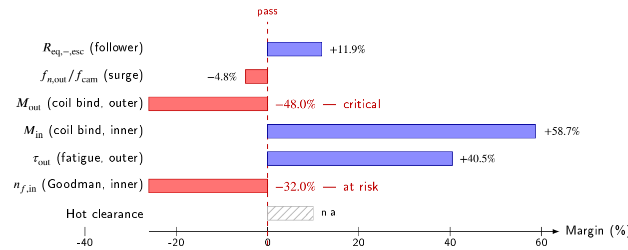

Figure 2: Tornado plot of the sensitivity of each preventive-barrier indicator to its dominant input under a ±10 % perturbation. Two indicators (in red) violate their pass criteria already in the baseline configuration and are identified as critical preventive barriers of the AP 1.8 valve train at 10 000 rpm: the outer-spring coil-bind margin M_{out} (−48.0%) and the inner-spring fatigue Goodman factor of safety $n_{f,in}$ (−32.0%). The outer-spring natural-frequency margin $f_{n,out}/f_{cam,max}$ (−4.8%) sits just below the conventional 4× rule-of-thumb threshold and is treated separately as a residual concern.

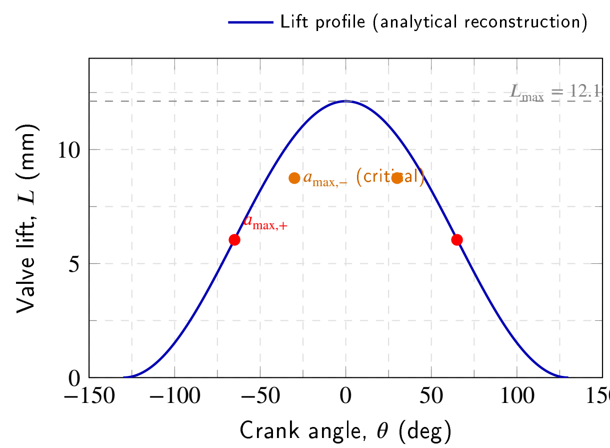

Figure 3: Schematic position of the two critical points of the cam profile of the AP 1.8 high-performance head: positive-acceleration peak ($a_{max,+}$ = 14 544 m s^{-2} at L_+ = 6.04 mm) and deceleration peak ($a_{max,-}$ = 20 362 m s^{-2} at L_- = 8.747 mm). The full lift profile is generated from the cam sizing of the intake valve.

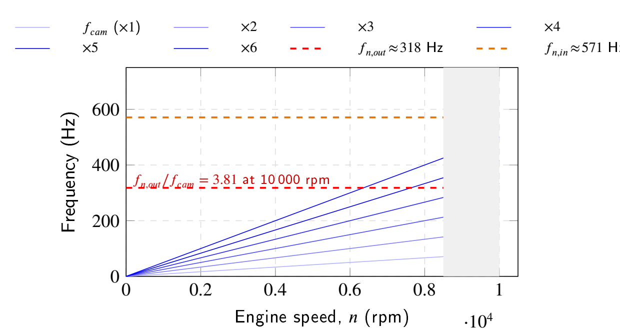

Figure 4: Spring natural frequencies $f_{n,out}$ and $f_{n,in}$ compared with the first six cam harmonics ($n = 1, \ldots, 6$) across the engine-speed range up to 10 000 rpm. The shaded region marks the operating regime; the dashed horizontal lines mark $f_{n,out} \approx 318$ Hz and $f_{n,in} \approx 571$ Hz.